EF101-P04: Project Resources and Team Roles

What the project is about

EF101-P04 measures how one laptop behaves under idle, CPU, memory, storage, mixed and GPU workloads, and finds which part limits speed in each case. The test machine is an HP EliteBook 850 G5 (Core i5-8250U, 4 cores / 8 threads, 7.8 GB usable RAM, 256 GB Samsung SSD, Intel UHD 620, Windows 10 Pro). A PowerShell script logs nine Windows counters every 2 seconds, HWiNFO adds clock, temperature and power, and Python re-checks the data and draws the graphs. Three students share the work; every member must still be able to explain the whole system in the individual viva.

Team

Member	Roll no.	Role
Nabia Shamsi	EE-001	Member
Areeza Rehan	EE-010	Member
Abdul Muqeet	EE-038	Member

Tools and equipment we used

The project uses one laptop, four built-in Windows tools, one free sensor program and Python for analysis. No database, cloud service or internet connection is needed; only the video in the mixed and GPU tests uses the internet.

Item	Category	How we used it
HP EliteBook 850 G5 (i5-8250U, 7.8 GB RAM, 256 GB SSD, Intel UHD 620)	Hardware	The single machine under test
Windows 10 Pro	Operating system	Runs the tests and keeps the performance counters
PowerShell 5.1: Get-Counter, Start-Job, Compress-Archive, Export-Csv	Built-in tool	Reads 9 counters every 2 s; creates the CPU, RAM and storage loads; writes CSV rows
Log-Test.ps1	Our script (Bonus B)	Runs one logged test and saves one row every 2 seconds to Raw_Logs
HWiNFO64	Free tool	Clock speed, temperature, power and throttling flags (Bonus A)
Task Manager	Built-in tool	Visual check and screenshots during the RAM, storage and mixed tests
Browser (5 tabs) and full-screen video	Workload	Mixed workload and light GPU workload
Excel	Software	Opens CSV files for quick checks (row count, empty cells, time order)
Python with pandas and matplotlib: analyze_logs.py	Software	Quality check, summary tables, re-check of the numbers; graphs 00 to 11 and the report were built with Python
Project folder EF101-P04 (Raw_Logs, Cleaned_Data, Graphs, Screenshots, Report, Scripts, Practice_Runs)	Data	Evidence for submission; raw logs are never edited
Complete Theory Guide, Concepts and Viva Guide, Presentation and Demo Guide	Study material	Revision for the viva and the live demo
Microsoft Learn (Get-Counter, Start-Job, Export-Csv), HWiNFO, Intel ARK	References	Counter definitions, tool documentation, processor limits
Claude (Anthropic)	AI assistant	Helped draft the scripts and the first draft of the report; the group ran every test and checked every number (disclosed in report section 12)

Who did what

All three members planned and ran every test together on the same laptop. After the tests the work was shared as below, so that each member owns a part of the project and can explain it.

	Nabia Shamsi	Areeza Rehan	Abdul Muqeet
Main work	Data cleaning and quality check; summary tables; graphs 00 to 11; bottleneck analysis; the report; requirements check	Specifications; HWiNFO logging; wrote and maintains Log-Test.ps1 (Bonus B); thermal-throttling analysis (Bonus A); GPU test results	Checked every raw CSV (rows, empty cells, time order); kept Practice_Runs; idle baseline and mixed workload set-up; runs the live demo
Tests	Ran all tests together with the group	Ran all tests together with the group	Ran all tests together with the group
Gave the project	Cleaned CSVs; QC report; Graphs 00 to 11; project report (.docx and .pdf)	Specs file; HWiNFO session files; Scripts/Log-Test.ps1; screenshots 01, 03, 04, 08	Data-check notes; Practice_Runs; screenshots 02, 06
In the presentation	Introduction; architecture; results, bottlenecks and graphs 04, 06, 07, 08; verdict	Laptop and specs; HWiNFO; CPU test and throttling (graphs 02, 03); GPU result	How the tests were run; the script; live demo; data checks
Will explain	Bottleneck per workload; utilization vs performance; why a laptop is slow at low CPU; pagefile; storage and cache; limitations	Cores vs threads; why HWiNFO; counters and try/catch; no throttling; GPU test; the invalid first CPU run and its fix	Two windows, logging first; the workloads; idle baseline and its short rest; data checks; why raw files are never edited

Step by step

1. The group records the specifications, Areeza writes Log-Test.ps1, and HWiNFO is started before each load that needs sensor data.
2. All three run the tests together (idle, CPU, cooldown, RAM, storage, mixed, GPU). In every test the logger starts first, then the workload (the first CPU run failed because this order was reversed).
3. Raw CSV files go into Raw_Logs and are never edited; Abdul checks row count, empty cells and time order of every file.
4. Nabia cleans the data, re-checks the numbers with analyze_logs.py and draws the graphs into their own folders.
5. Nabia writes the report and the requirements check; all three review the graphs and the report before submission.

Tests were run on 13 Oct (practice), 15 Oct (idle, RAM, storage, mixed), 18 Oct (valid CPU run and cooldown) and 21 Oct (GPU).